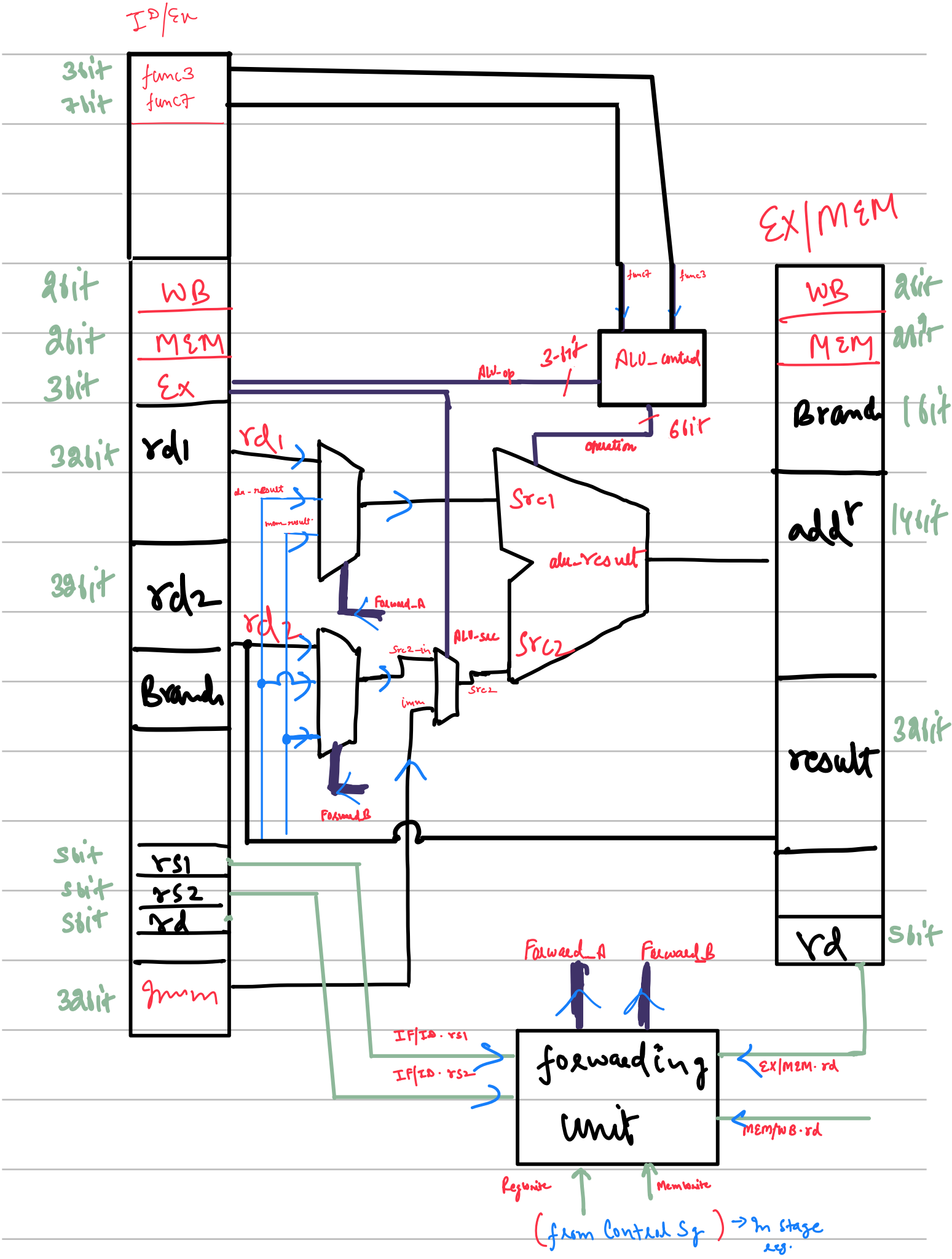
ID/EX
3bit
7bit
func3
func7
2bit
WB
2bit
MEM
3bit
EX
32bit
rd1
32bit
rd2
Branch
5bit
rs1
5bit
rs2
5bit
rd
32bit
imm
EX|MEM
WB
2bit
MEM
2bit
Branch
1bit
addr
14bit
result
32bit
rd
5bit
func7
func3
ALU_control
3-bit
ALU-op
6bit
operation
rd1
alu-result
mem-result
Forward_A
rd2
Forward_B
Src1
alu-result
Src2
ALU-sec
Src2-in
Src2
imm
Forward_A
Forward_B
IF/ID.rs1
IF/ID.rs2
forwarding unit
EX|MEM.rd
MEM/WB.rd
Regwrite
Memwrite
(from Control Sig) → in stage reg.